

User Manual

Automated Data Dictionary for MSA ERP System

1. Purpose of the System

This system automates the creation, validation, storage, and publishing of a centralized **Data Dictionary** for the MSA ERP ecosystem. It ensures that all metadata submitted by users is:

- Reviewed and approved by admins
- Stored securely in SQL databases
- Ingested into Microsoft Fabric
- Published as a live, queryable data dictionary
- Automatically refreshed in downstream services

The goal is to eliminate manual documentation, improve data governance, and ensure one trusted source of truth.

2. User Roles and Responsibilities

2.1 Standard User

- Submits data dictionary entries through the Power App form
- Views published data dictionary through FreshService
- Cannot directly modify approved records

Power Apps User Interface:

The screenshot displays the Power Apps 'UserInterface (Editing)' form. At the top, a red header bar contains the 'WISCONSIN' logo, the text 'Power Apps | UserInterface (Editing)', and environment information 'Environment: UW-Madison (default)'. Below the header, a green notification bar states 'Feedback submitted for review.' The main form area features a 'Filter by Table Name' dropdown set to 'vw_ActualData' and a 'Search by Column Name' input field. A list of columns is shown on the left, including 'ActualInvoiceAmount_vw_ActualData', 'ActualHours_vw_ActualData', 'ActualAmount_vw_ActualData', 'ActualAmount_FeeMgt_vw_ActualData', 'ActualAmount_Expense_vw_ActualData', and 'ActualAmount_Labor_vw_ActualData'. The right side of the form contains a 'Description' field with the text 'This column stores the monetary amount related to actualinvoiceamount.', a 'Feedback' section with a text area containing 'This is a Test Description', and a 'Notes' section with another text area. A blue 'Save' button is located at the bottom right of the form.

2.2 Admin

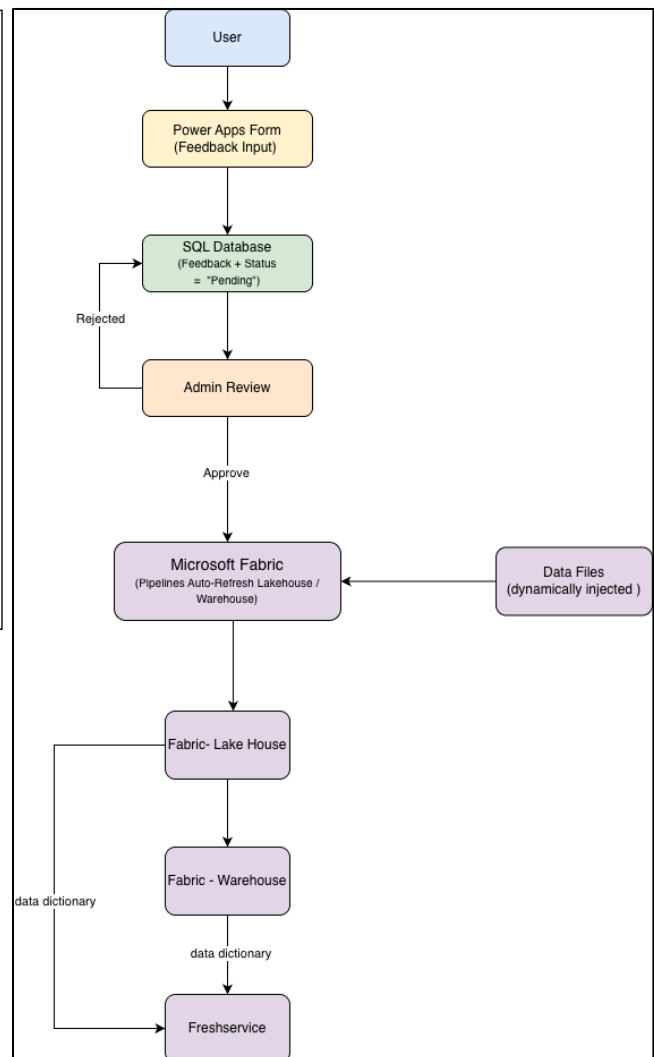
- Reviews all user submissions
- Approves or rejects entries
- Manages data quality and schema validation
- Oversees ingestion workflows

2.3 System Services (Automated)

- SQL Server for secure data storage
- GitHub API for file transfer
- Microsoft Fabric for lakehouse and warehouse processing
- FreshService for end-user access

3. High-Level System Workflow

1. User submits metadata using Power App
2. Admin reviews and approves the submission
3. Approved data is stored in SQL Server
4. Data is exported as structured files
5. Files are transferred using HTTPS API
6. Data is injected into Fabric Lakehouse
7. Data is curated in Fabric Warehouse
8. Data Dictionary is published
9. FreshService displays the live dictionary.



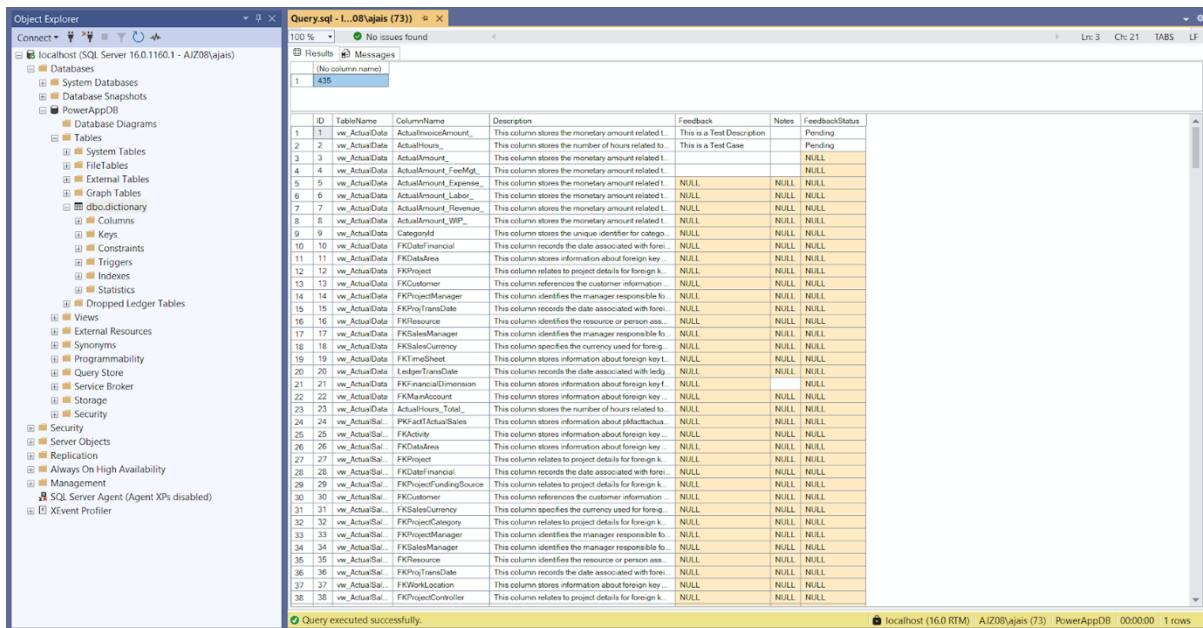
4. Detailed Operational Workflow

4.1 Data Submission via Power App

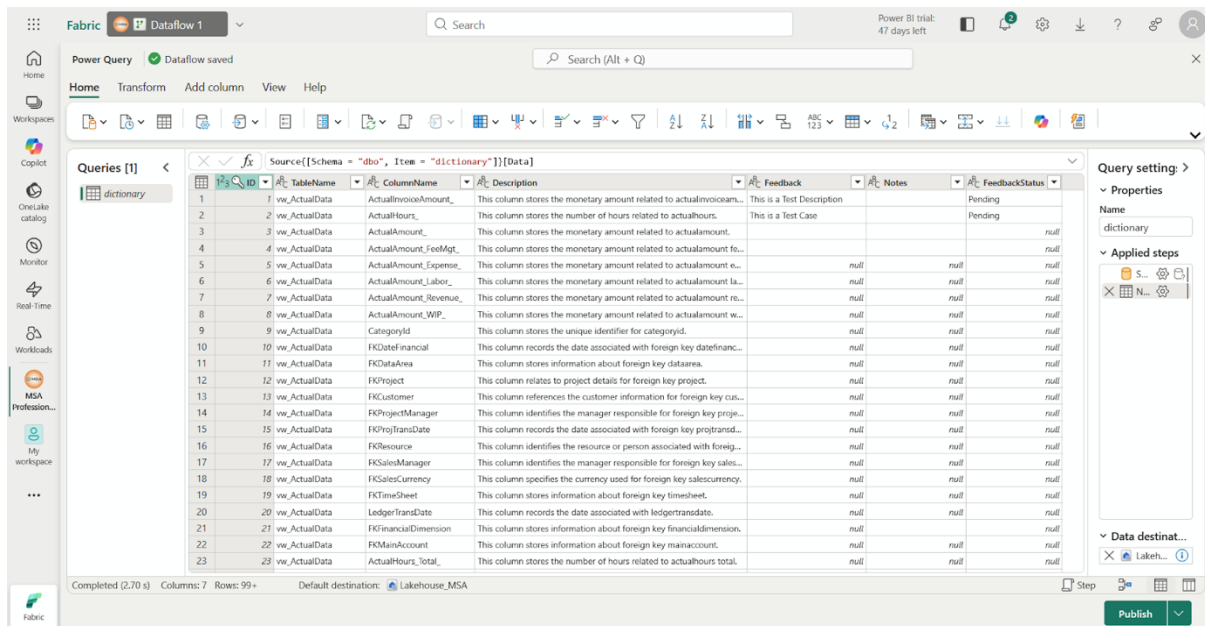
1. User logs into the Power App Form.
2. User enters:
 - Table name
 - Column name
 - Data type
 - Business definition
 - Data owner
 - Sensitivity level
3. User submits the form.
4. Submission status becomes **Pending Approval**.

The screenshot shows the Power Apps 'UserInterface (Editing)' screen. At the top, there is a red header bar with the Wisconsin Department of Business Services logo and the text 'Power Apps | UserInterface (Editing)'. Below the header, a green notification bar states 'Feedback submitted for review.' The main content area is divided into two sections. On the left, under 'Filter by Table Name :', a dropdown menu is set to 'vw_ActualData'. Below this, a list of columns is displayed, each with a right-pointing chevron: 'ActualInvoiceAmount_vw_ActualData', 'ActualHours_vw_ActualData', 'ActualAmount_vw_ActualData', 'ActualAmount_FeeMgt_vw_ActualData', 'ActualAmount_Expense_vw_ActualData', and 'ActualAmount_Labor_vw_ActualData'. On the right, the 'Description' field contains the text 'This column stores the monetary amount related to actualinvoiceamount.' Below this is a 'Feedback' section with a text input field containing 'This is a Test Description'. Further down is a 'Notes' section with a larger text input field. At the bottom right of the main content area is a blue 'Save' button. The entire interface is set against a light gray grid background.

Power Apps User Interface



SQL Database Feedback Status Changes to “Pending”:

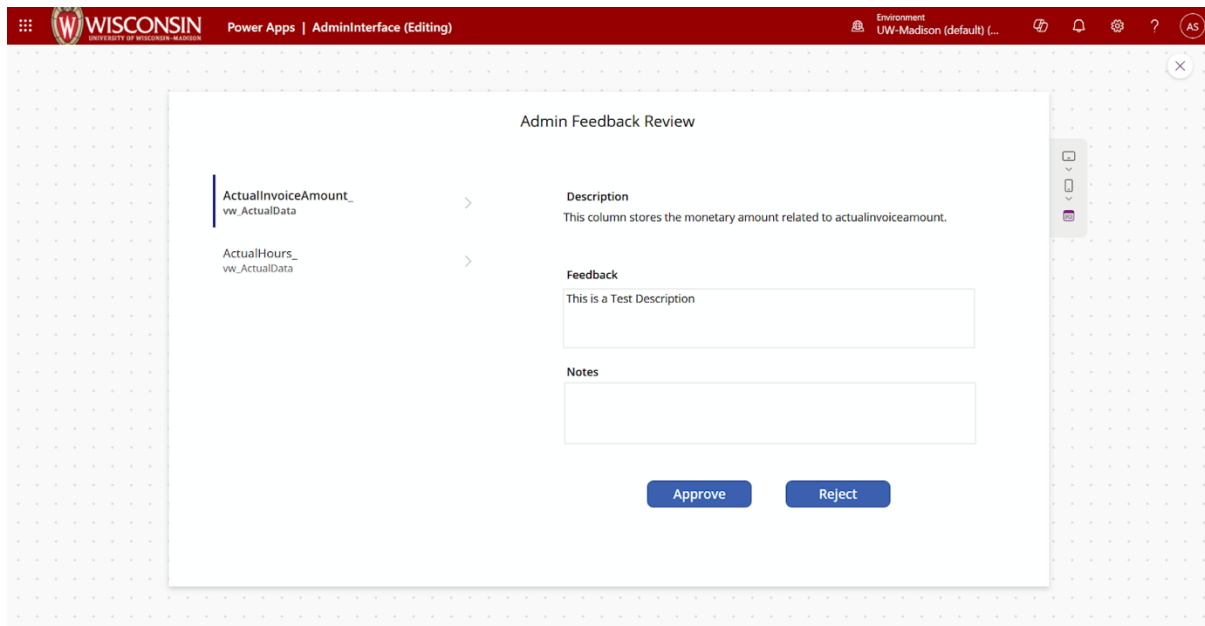


Fabric Data Flow Feedback Status Changes to “Pending”

4.2 Admin Approval Process

1. Admin logs into the Admin Panel.
2. Admin reviews each submission:
 - o Checks for naming consistency

- Validates definitions
- Verifies data ownership
- 3. Admin selects:
 - **Approve** to store in SQL Server
 - **Reject** to return to the user with comments
- 4. Approved records are written to the SQL Database.



Power Apps Admin Interface

4.3 Data Storage in SQL Server

- Approved metadata is stored in structured relational tables.
- Each record includes:
 - Submission ID
 - Approval timestamp
 - Admin reviewer
 - Version number
- SQL Server acts as the system of record.

Object Explorer

Connect - 100% - No issues found

localhost (SQL Server 16.0.1160.1 - AZ208\ajais)

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.dictionariy

Columns

Keys

Constraints

Triggers

Indexes

Statistics

Dropped Ledger Tables

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Server Objects

Replication

Always On High Availability

Management

SQL Server Agent (Agent XPs disabled)

XEvent Profiler

Query: sql - L.08\ajais (731)

Results

Messages

[No column names]

1 435

ID	TableName	ColumnName	Description	Feedback	Notes	FeedbackStatus
1	vw_ActualData	ActualInvoiceAmount_	This is a Test Description	NULL		Approved
2	vw_ActualData	ActualHours_	This is a Test Case	NULL		Approved
3	vw_ActualData	ActualAmount	This column stores the monetary amount related to actualamount			NULL
4	vw_ActualData	ActualAmount_FeeMgt	This column stores the monetary amount related to actualamount feeMgt			NULL
5	vw_ActualData	ActualAmount_Expense	This column stores the monetary amount related to actualamount expense	NULL	NULL	NULL
6	vw_ActualData	ActualAmount_Labor	This column stores the monetary amount related to actualamount labor	NULL	NULL	NULL
7	vw_ActualData	ActualAmount_Revenue	This column stores the monetary amount related to actualamount revenue	NULL	NULL	NULL
8	vw_ActualData	ActualAmount_WIP	This column stores the monetary amount related to actualamount wip	NULL	NULL	NULL
9	vw_ActualData	CategoryId	This column stores the unique identifier for categoryid	NULL	NULL	NULL
10	vw_ActualData	FKDateFinancial	This column records the date associated with foreign key datefinancial	NULL	NULL	NULL
11	vw_ActualData	FKDataArea	This column stores information about foreign key dataarea	NULL	NULL	NULL
12	vw_ActualData	FKProject	This column relates to project details for foreign key project	NULL	NULL	NULL
13	vw_ActualData	FKCustomer	This column references the customer information for foreign key customer	NULL	NULL	NULL
14	vw_ActualData	FKProjectManager	This column identifies the manager responsible for foreign key projectmanager	NULL	NULL	NULL
15	vw_ActualData	FKProjTransDate	This column records the date associated with foreign key projtransdate	NULL	NULL	NULL
16	vw_ActualData	FKResource	This column identifies the resource or person associated with foreign key resource	NULL	NULL	NULL
17	vw_ActualData	FKSalesManager	This column identifies the manager responsible for foreign key salesmanager	NULL	NULL	NULL
18	vw_ActualData	FKSalesCurrency	This column specifies the currency used for foreign key salescurrency	NULL	NULL	NULL
19	vw_ActualData	FKTimeSheet	This column stores information about foreign key timesheet	NULL	NULL	NULL
20	vw_ActualData	LedgerTransDate	This column records the date associated with ledgertransdate	NULL	NULL	NULL
21	vw_ActualData	FKFinancialDimension	This column stores information about foreign key financialdimension	NULL	NULL	NULL
22	vw_ActualData	FKMainAccount	This column stores information about foreign key mainaccount	NULL	NULL	NULL
23	vw_ActualData	ActualHours_Total	This column stores the number of hours related to actualhours total	NULL	NULL	NULL
24	vw_ActualSal	PKActActualSales	This column stores information about pkactactualsales	NULL	NULL	NULL
25	vw_ActualSal	FKActivity	This column stores information about foreign key activity	NULL	NULL	NULL
26	vw_ActualSal	FKDataArea	This column stores information about foreign key dataarea	NULL	NULL	NULL
27	vw_ActualSal	FKProject	This column relates to project details for foreign key project	NULL	NULL	NULL
28	vw_ActualSal	FKDateFinancial	This column records the date associated with foreign key datefinancial	NULL	NULL	NULL
29	vw_ActualSal	FKProjectFundingSource	This column relates to project details for foreign key projectfundingsource	NULL	NULL	NULL
30	vw_ActualSal	FKCustomer	This column references the customer information for foreign key customer	NULL	NULL	NULL
31	vw_ActualSal	FKSalesCurrency	This column specifies the currency used for foreign key salescurrency	NULL	NULL	NULL
32	vw_ActualSal	FKProjectCategory	This column relates to project details for foreign key projectcategory	NULL	NULL	NULL
33	vw_ActualSal	FKProjectManager	This column identifies the manager responsible for foreign key projectmanager	NULL	NULL	NULL

Query executed successfully.

localhost (16.0 RTM) AZ208\ajais (73) PowerAppDB 00:00:00 1 rows

SQL Database Feedback Status Changes to “Approved”, if admin approves the changes and then Description gets updated with the new Feedback, else if “Rejected” everything resets back to NULL and Feedback does not change.

Fabric

Dataflow 1

Power Query

Dataflow saved

Search

Search (Alt + Q)

Home Transform Add column View Help

Queries [1]

dictionary

Source: [Schema = "dbo", Item = "dictionary"] [Data]

ID	TableName	ColumnName	Description	Feedback	Notes	FeedbackStatus
1	vw_ActualData	ActualInvoiceAmount_	This is a Test Description	null		Approved
2	vw_ActualData	ActualHours_	This is a Test Case	null		Approved
3	vw_ActualData	ActualAmount	This column stores the monetary amount related to actualamo...			null
4	vw_ActualData	ActualAmount_FeeMgt	This column stores the monetary amount related to actualamo...			null
5	vw_ActualData	ActualAmount_Expense	This column stores the monetary amount related to actualamo...	null		null
6	vw_ActualData	ActualAmount_Labor	This column stores the monetary amount related to actualamo...	null		null
7	vw_ActualData	ActualAmount_Revenue	This column stores the monetary amount related to actualamo...	null		null
8	vw_ActualData	ActualAmount_WIP	This column stores the monetary amount related to actualamo...	null		null
9	vw_ActualData	CategoryId	This column stores the unique identifier for categoryid	null		null
10	vw_ActualData	FKDateFinancial	This column records the date associated with foreign key datefi...	null		null
11	vw_ActualData	FKDataArea	This column stores information about foreign key dataarea	null		null
12	vw_ActualData	FKProject	This column relates to project details for foreign key project	null		null
13	vw_ActualData	FKCustomer	This column references the customer information for foreign ke...	null		null
14	vw_ActualData	FKProjectManager	This column identifies the manager responsible for foreign key ...	null		null
15	vw_ActualData	FKProjTransDate	This column records the date associated with foreign key projr...	null		null
16	vw_ActualData	FKResource	This column identifies the resource or person associated with L...	null		null
17	vw_ActualData	FKSalesManager	This column identifies the manager responsible for foreign key ...	null		null
18	vw_ActualData	FKSalesCurrency	This column specifies the currency used for foreign key salescur...	null		null
19	vw_ActualData	FKTimeSheet	This column stores information about foreign key timesheet	null		null
20	vw_ActualData	LedgerTransDate	This column records the date associated with ledgertransdate	null		null
21	vw_ActualData	FKFinancialDimension	This column stores information about foreign key financialdime...	null		null
22	vw_ActualData	FKMainAccount	This column stores information about foreign key mainaccount	null		null
23	vw_ActualData	ActualHours_Total	This column stores the number of hours related to actualhours ...	null		null

Completed (2.22 s) Columns: 7 Rows: 99+ Default destination: Lakehouse_MSA

Query setting: >

Properties

Name

dictionary

Applied steps

Data destination

Lakehouse_MSA

Publish

Fabric DataFlow also gets Updated

4.4 Automated Data File Generation

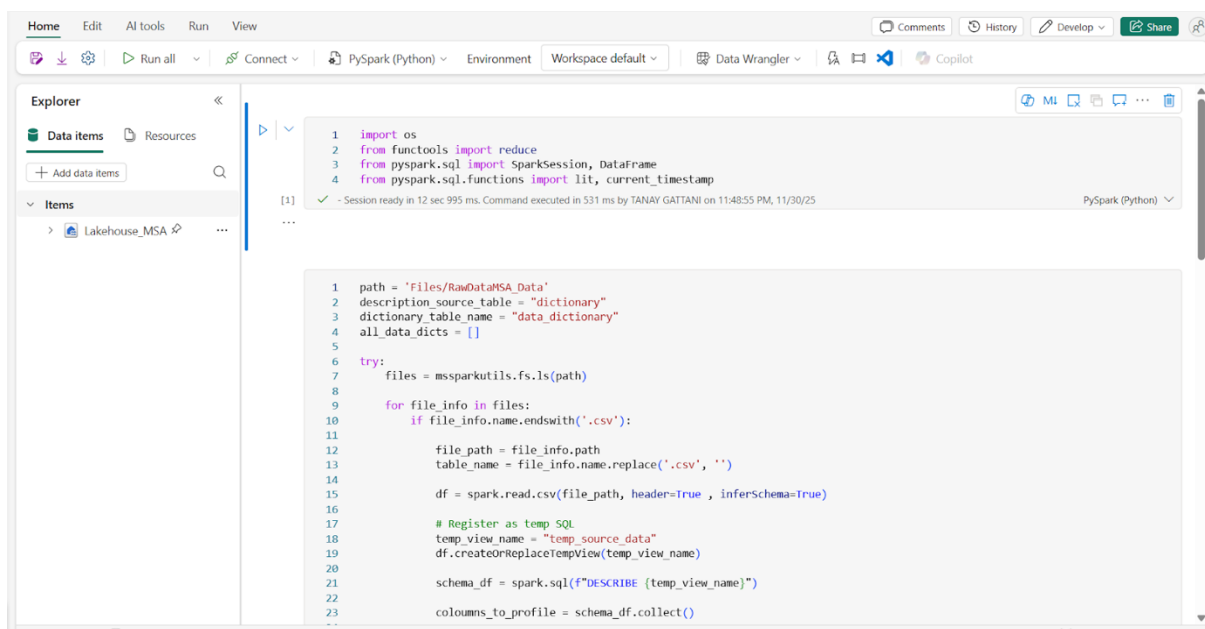
- A scheduled GitHub-based service:
 - Extracts approved SQL records
 - Converts them into structured data files (CSV/JSON)
- Files are securely packaged for transfer.

4.5 Secure Data Transfer (HTTPS API)

- Data files are transmitted via HTTPS API.
- Transfer includes:
 - Authentication tokens
 - Encrypted payload
- Transfer success or failure is logged automatically.

4.6 Data Injection into Fabric Lakehouse

- Incoming files are landed into the Fabric Lakehouse.
- Raw metadata is stored in staging layers.
- Schema validation and formatting occur automatically.



The screenshot displays the Databricks IDE interface. On the left, the 'Explorer' sidebar shows a project named 'Lakehouse_MSA'. The main workspace contains a PySpark script with the following code:

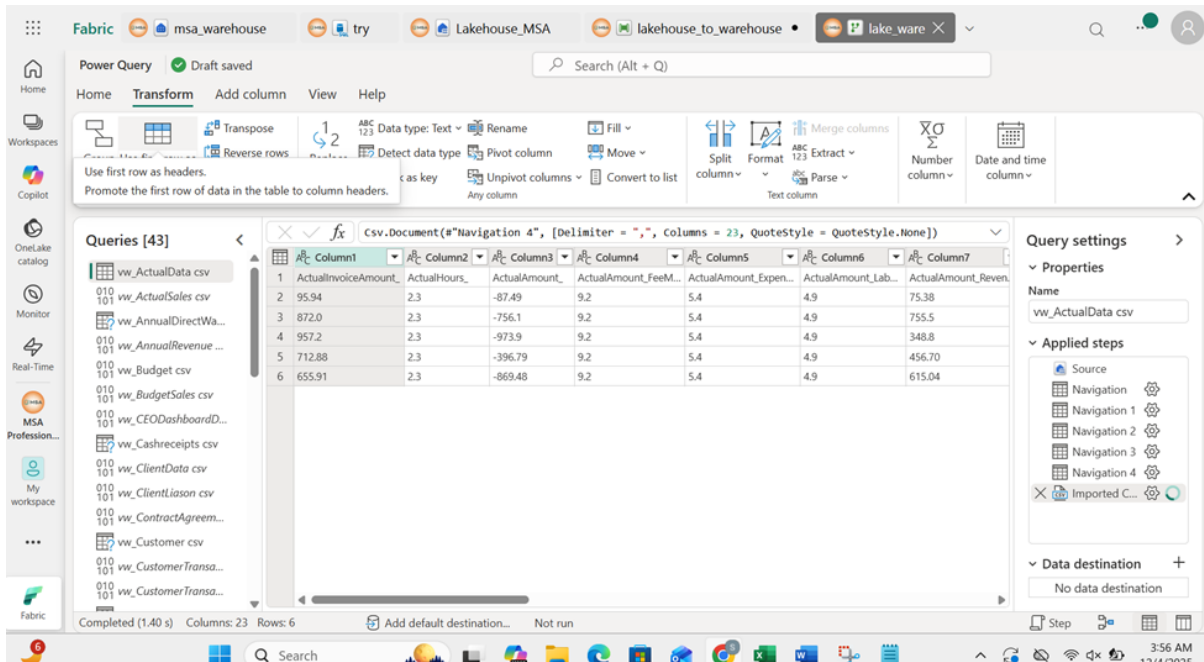
```
1 import os
2 from functools import reduce
3 from pyspark.sql import SparkSession, DataFrame
4 from pyspark.sql.functions import lit, current_timestamp

[1] ✓ - Session ready in 12 sec 995 ms. Command executed in 531 ms by TANAY GATTANI on 11:48:55 PM, 11/30/25

1 path = 'Files/RawDataMSA.Data'
2 description_source_table = "dictionary"
3 dictionary_table_name = "data_dictionary"
4 all_data_dicts = []
5
6 try:
7     files = mssparkutils.fs.ls(path)
8
9     for file_info in files:
10         if file_info.name.endswith('.csv'):
11
12             file_path = file_info.path
13             table_name = file_info.name.replace('.csv', '')
14
15             df = spark.read.csv(file_path, header=True, inferSchema=True)
16
17             # Register as temp SQL
18             temp_view_name = "temp_source_data"
19             df.createOrReplaceTempView(temp_view_name)
20
21             schema_df = spark.sql(f"DESCRIBE {temp_view_name}")
22
23             columns_to_profile = schema_df.collect()
```

4.7 Data Processing in Fabric Warehouse

- Cleaned data is transformed into:
 - Dimension tables
 - Dictionary reference tables
- Versioning and historical tracking are enabled here.
- This layer becomes the authoritative **Data Dictionary Store**.



4.8 Data Dictionary Publishing

- The curated data dictionary is exposed as:
 - Structured views
 - Searchable dictionary tables
- Data is refreshed automatically based on ingestion schedules.

data_dictionary

Showing 450 rows

Table view

ABC	Table_name	ABC	Column_name	ABC	Data_Type	ABC	Description	12L	Null_Count	12L	Not_Null_Cou...	12L	Total_Rows	12	Null_Percenta...	12L	Unique_Values	ABC	Sample_Value	Generated_D...
	vw_ActualData		ActualAmount_		double		This column store...	0	5	5	5	5	0	5	0	5		-973.9		2025-12-05T17:01...
	vw_ActualData		FKDataArea		string		This column store...	0	5	5	5	0	1	1	1	1		egt		2025-12-05T17:01...
	vw_ActualData		CategoryId		string		This column store...	0	5	5	5	0	1	1	1	1		Eyzas		2025-12-05T17:01...
	vw_ActualData		ActualInvoiceAmo...		double		This is a test case	0	5	5	5	0	5	5	0	5		957.2		2025-12-05T17:01...
	vw_ActualData		FKDateFinancial		string		This column recor...	0	5	5	5	0	1	1	1	1		fjx92071032		2025-12-05T17:01...
	vw_ActualData		ActualAmount_WIP_		double		This column store...	0	5	5	5	0	1	1	1	1		8.7		2025-12-05T17:01...
	vw_ActualData		ActualAmount_Ex...		double		This column store...	0	5	5	5	0	1	1	1	1		5.4		2025-12-05T17:01...
	vw_ActualData		FKProjTransDate		int		This column recor...	0	5	5	5	0	5	5	0	5		77575864		2025-12-05T17:01...
	vw_ActualData		FKProject		string		This column relate...	0	5	5	5	0	4	4	4	4		vftq568628173.09...		2025-12-05T17:01...
	vw_ActualData		FKResource		bigint		This column identi...	0	5	5	5	0	4	4	4	4		9578452075		2025-12-05T17:01...
	vw_ActualData		ActualAmount_Fe...		double		This is a test case	0	5	5	5	0	1	1	1	1		9.2		2025-12-05T17:01...
	vw_ActualData		ActualAmount_La...		double		This column store...	0	5	5	5	0	1	1	1	1		4.9		2025-12-05T17:01...
	vw_ActualData		FKCustomer		string		This column refere...	0	5	5	5	0	1	1	1	1		Parker Carter		2025-12-05T17:01...
	vw_ActualData		FKSalesManager		bigint		This column identi...	0	5	5	5	0	1	1	1	1		3785940984		2025-12-05T17:01...
	vw_ActualData		ActualHours		double		This column store...	0	5	5	5	0	1	1	1	1		2.3		2025-12-05T17:01...
	vw_ActualData		ActualAmount_Re...		double		This column store...	0	5	5	5	0	5	5	0	5		755.5		2025-12-05T17:01...
	vw_ActualData		FKSalesCurrency		string		This column specif...	0	5	5	5	0	1	1	1	1		RT5		2025-12-05T17:01...
	vw_ActualData		ActualHours_Tot...		double		This column store...	0	5	5	5	0	1	1	1	1		9.8		2025-12-05T17:01...
	vw_ActualData		FKMainAccount		bigint		This column store...	0	5	5	5	0	1	1	1	1		4267744751		2025-12-05T17:01...
	vw_ActualData		FKFinancialDimen...		string		This column store...	0	5	5	5	0	4	4	4	4		YMJ0432366287		2025-12-05T17:01...
	vw_ActualData		LedgerTransDate		date		This column recor...	0	5	5	5	0	1	1	1	1		6153-03-03		2025-12-05T17:01...

Succeeded (17 sec 447 ms)

Columns 11 Rows 450

Data Dictionary

4.9 FreshService Integration

- FreshService pulls from the Data Dictionary service.
- End users can:
 - Search column definitions
 - View ownership and sensitivity
 - Track version history
- No manual refresh is required.

5. User Instructions

5.1 How to Submit a New Data Entry

1. Open the Power App.
2. Select **New Dictionary Entry**.
3. Fill in all required metadata fields.
4. Click **Submit**.
5. Wait for admin approval.
6. Monitor status in the **My Submissions** view.

5.2 How to View the Data Dictionary

1. Open FreshService.
2. Navigate to **Data Dictionary Portal**.
3. Use filters by:
 - Table name
 - Column name
 - Business domain
4. Click a field to view:
 - Definition
 - Owner
 - Sensitivity
 - Last updated timestamp

6. Admin Instructions

6.1 Review and Approval

1. Open Admin Console.

2. Navigate to **Pending Submissions**.
3. Select a submission.
4. Validate:
 - Completeness
 - Business meaning
 - Compliance
5. Choose **Approve** or **Reject**.
6. Add optional review notes.

6.2 Monitoring Data Pipelines

Admins must regularly monitor:

- SQL ingestion jobs
- File export logs
- API transfer logs
- Fabric ingestion success messages

Error alerts are auto-generated for:

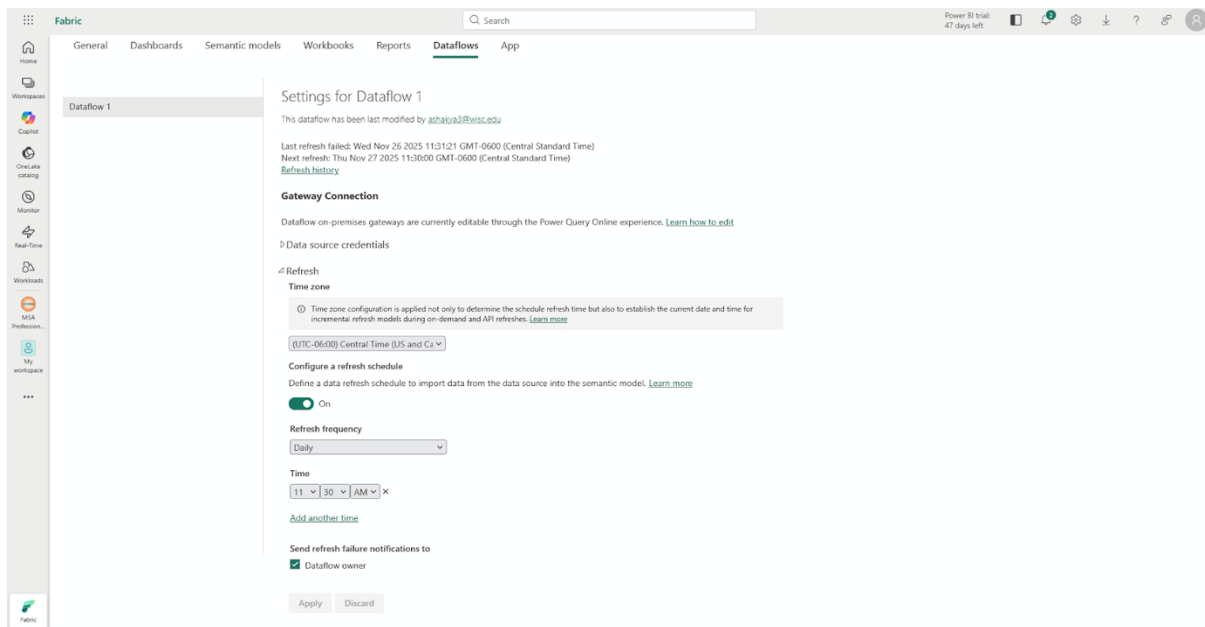
- Failed API calls
- File validation errors
- Warehouse load failures

7. Security and Compliance

- All API traffic is encrypted using HTTPS.
- Role-based access control is enforced at:
 - Power App
 - SQL Server
 - Fabric
 - FreshService
- Audit logs track:
 - Submissions
 - Approvals
 - Modifications
 - Access history

8. Data Refresh & Versioning

- Data refresh runs on scheduled intervals.
- Every update creates:
 - A new version entry
 - Historical records for traceability
- Users only see the latest approved version by default.



Fabric: From Data Flow Settings we can schedule a particular time for auto refresh in order to update changes from DataFlow to fabric database

The screenshot shows the Microsoft Fabric home page. The top navigation bar includes 'Fabric', a search bar, and a 'Power BI trial: 47 days left' message. Below the navigation bar, there's a section for 'MSA Professional...' with buttons for '+ New item', 'New folder', 'Import', and 'Migrate'. The main content area is a table listing various dataflows and pipelines. The table has columns for Name, Status, Type, Task, Owner, Refreshed, Next refresh, Endorsement, and Reset. The 'dataflow_msa' entry is highlighted with a red status icon.

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Reset
Dataflow		Dataflow Gen2	—	Ajai Shukya	11/26/2025, 12:34:35 PM	11/27/2025, 11:30:00 AM	—	Reset
dataflow_msa		Dataflow Gen...	—	Sai Keerthana Jangala	11/20/2025, 6:06:20 PM	—	—	—
Final_Nivdhba		Notebook	—	TANAY GATTANI	—	—	—	—
Lakehouse to Warehouse		Pipeline	—	Shubham Mehta	—	—	—	—
Lakehouse_MSA		Lakehouse	—	Shubham Mehta	—	—	—	—
Lakehouse_MSA		SQL analytics ...	—	Shubham Mehta	—	—	—	—
Let_give_it_a_try		Lakehouse	—	TANAY GATTANI	—	—	—	—
Let_give_it_a_try		SQL analytics ...	—	TANAY GATTANI	—	—	—	—
msa_warehouse		Warehouse	—	Sai Keerthana Jangala	—	—	—	—
Mypipeline		Pipeline	—	TANAY GATTANI	—	—	—	—
Parent_pipeline		Pipeline	—	Shubham Mehta	—	—	—	—
static_test		Pipeline	—	Shubham Mehta	—	—	—	—
test_1		Pipeline	—	Shubham Mehta	—	—	—	—

Fabric: Now after Data is Published from DataFlow: We can see refreshed and next refresh details from the home page itself

9. Error Handling & Troubleshooting

Issue	Likely Cause	Resolution
Submission not visible	Pending admin approval	Contact admin
Data not visible in FreshService	Pipeline delay	Wait for next refresh
Duplicate field entry	Naming conflict	Admin will reject
API failure	Network or token issue	System auto-retries

10. System Benefits

- Eliminates manual spreadsheet-based data dictionaries
- Enforces governance through admin approval
- Creates a single trusted metadata source
- Enables real-time discovery through FreshService
- Fully auditable and version-controlled